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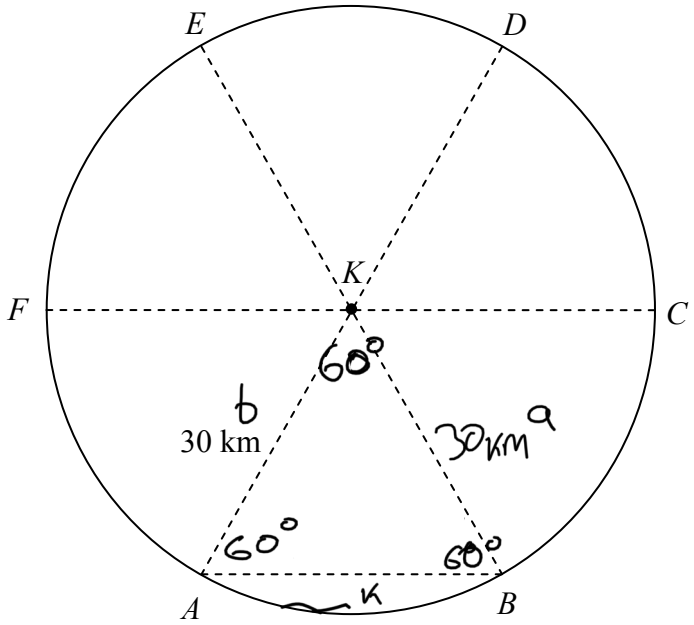
Question 8 **(75 marks)**

A search is begun for a buoy that has become detached from its mooring at sea. The area to be searched is a circle of radius 30 km from the last known position, K, of the buoy. The search area is divided into six equal sectors as indicated by the letters A, B, C, D, E and F.

$\pi = 180°$

$\frac{\pi}{3} = 60°$



(a) Fishing boats search the triangular area KAB.

(i) Find $|\angle BKA|$.

Answer: 60°

(ii) Find the area of the triangle KAB.

$\frac{1}{2}(30)(30)\sin 60°$

384.7114317 km^2

(iii) Write the area of the triangle *KAB* as a percentage of the area of the sector *KAB*.

$\pi(30)^2\left(\frac{60}{360^\circ}\right)$

$150\pi \Rightarrow 471.24\ km^2$

(iv) Use the cosine rule to find the length of $[AB]$.

$a^2 = b^2 + c^2 - 2bc\cos A$

$|AB| = 30^2 + 30^2 - 2(30)(30)\cos 60 = 400$

$400 + 900 - 900 = 900$

$\Rightarrow |AB| = \sqrt{900}$

$|AB| = 30\ km$

(v) What does your answer to **(iv)** above show about the triangle *KAB*?

It is equilateral as all sides are the same.

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(b) A helicopter took part in the search.

(i) The helicopter flew from the point F around the perimeter of the search area. What distance did the helicopter fly, correct to the nearest km?

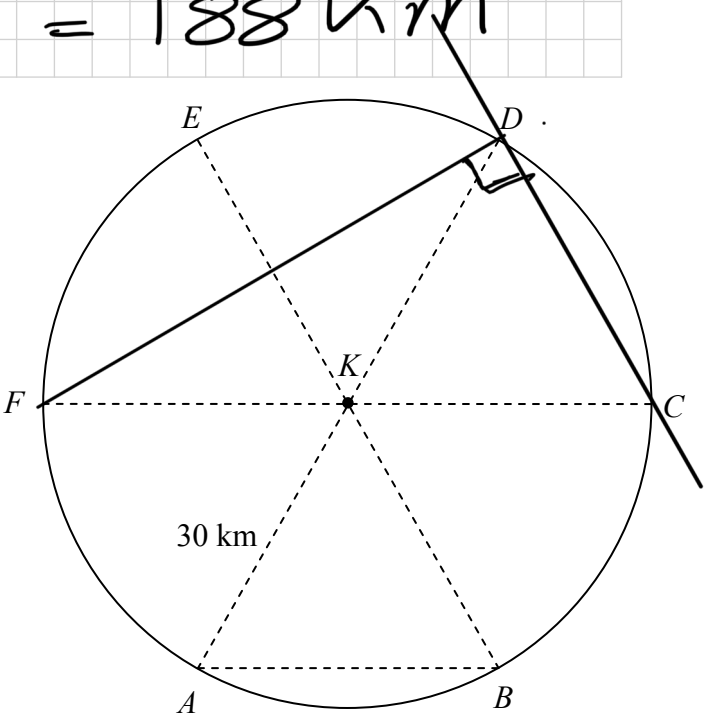
Circumference

$2\pi r$

$2\pi(30)$

$= 60\pi = 188$ km

(ii) The helicopter then flew in a straight line from F to D and from D on to C, also in a straight line. Draw the path of the helicopter on the diagram.



(iii) A theorem on your course can be used to find $|\angle FDC|$.
Write down $|\angle FDC|$ and state the theorem.

$|\angle FDC| =$ 90°

Statement of theorem:

(iv) The helicopter flew at a speed of 80 km/h. How long did it take to fly from F to D and on to C?

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(c) A lifeboat taking part in the search sailed, in a straight line, from the point K until it reached a point X, the midpoint of $[ED]$.

(i) Calculate $|KX|$.

(ii) The buoy was located at the point where the path KX, of the lifeboat, crossed the path FD of the helicopter. How far was the buoy from X?